

END-TO-END TECHNICAL DATA FLOW

# VortexEye

AI-Powered Indoor · Outdoor Navigation · Built for the 285 Million

● FULLY OPERATIONAL PROTOTYPE · 6,800+ LINES · 11 MODULES

● User Interaction

● Location Intelligence

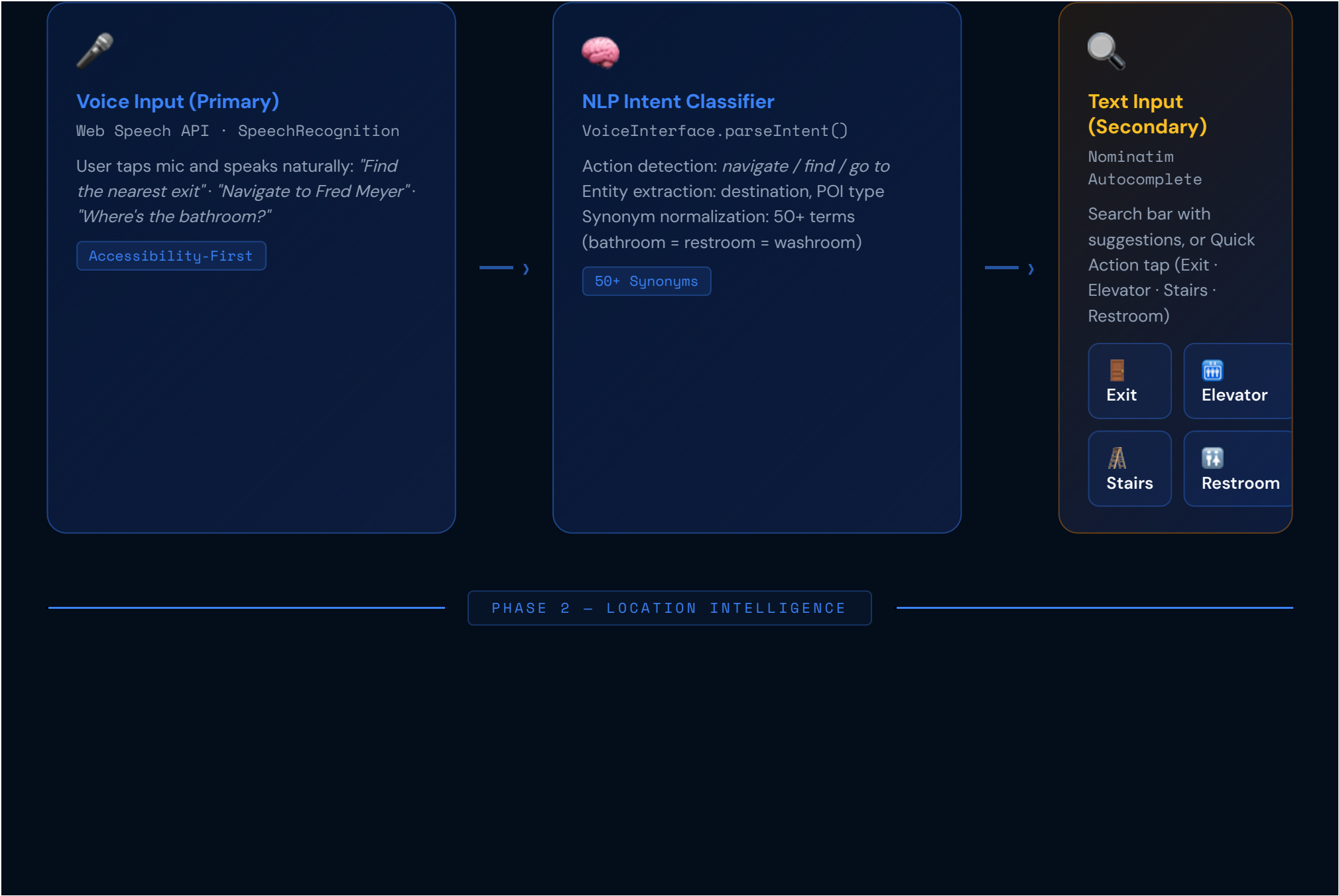
● Sensor Fusion

● Pathfinding & Routing

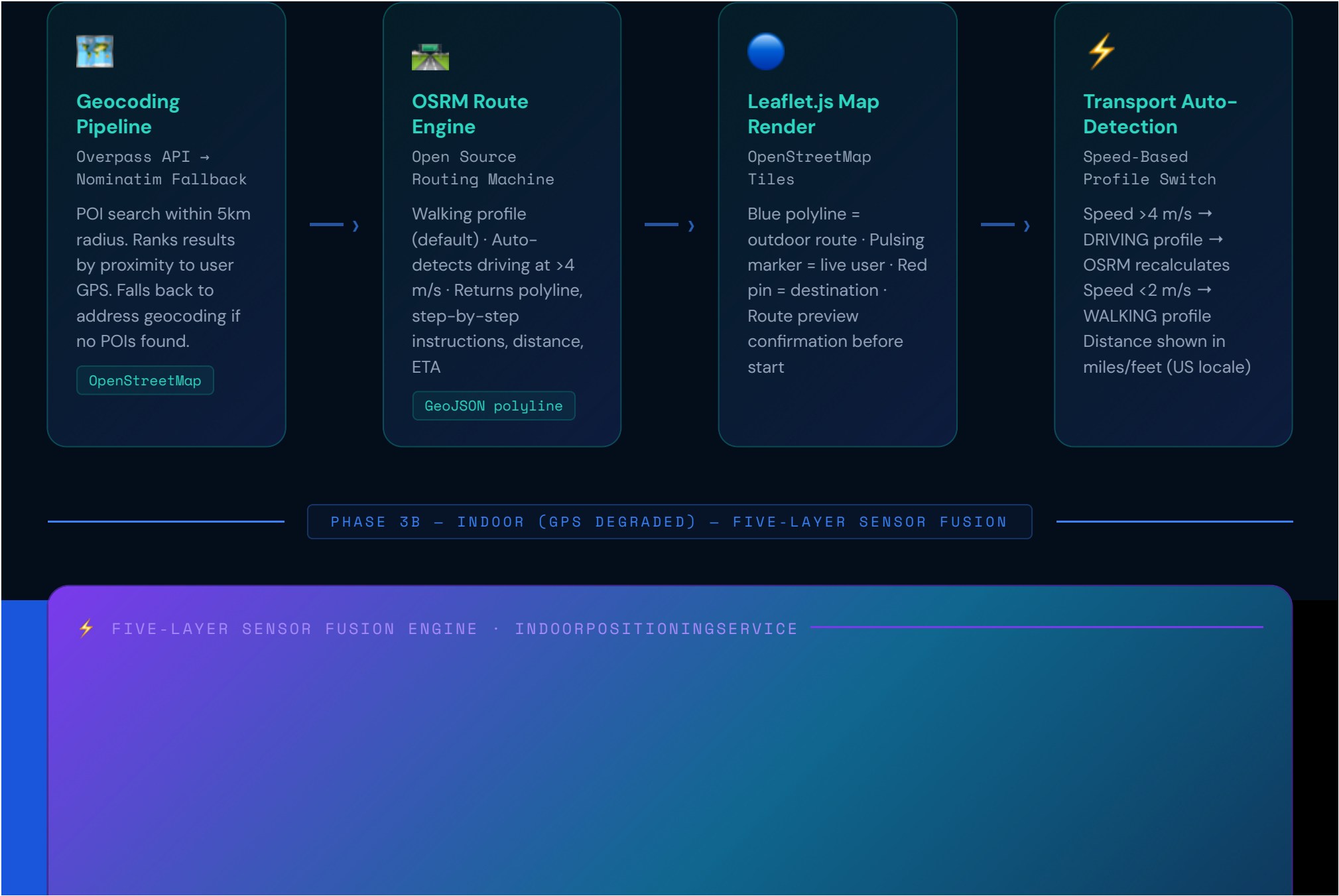
● Voice Guidance Output

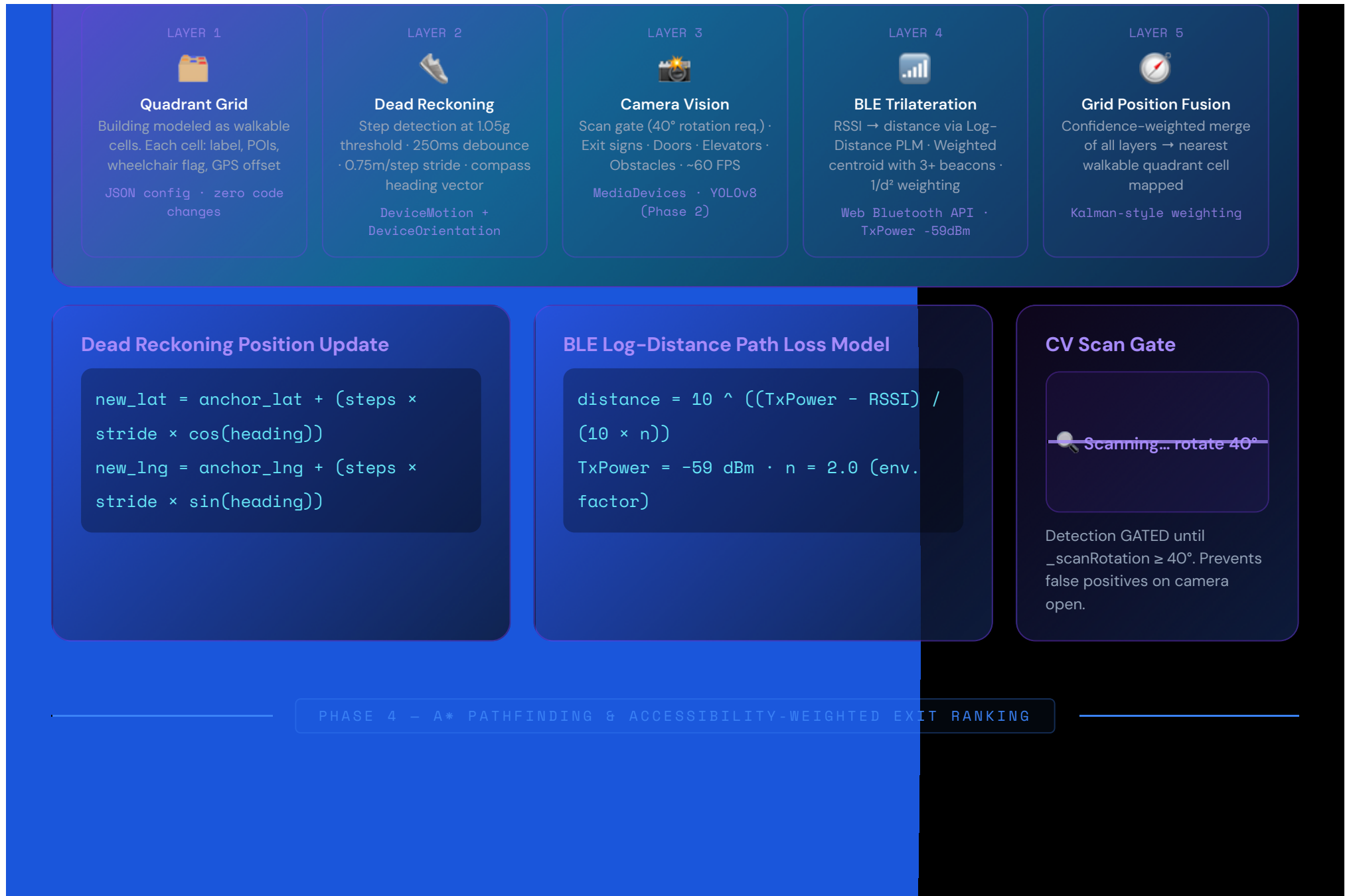
● Safety / Emergency

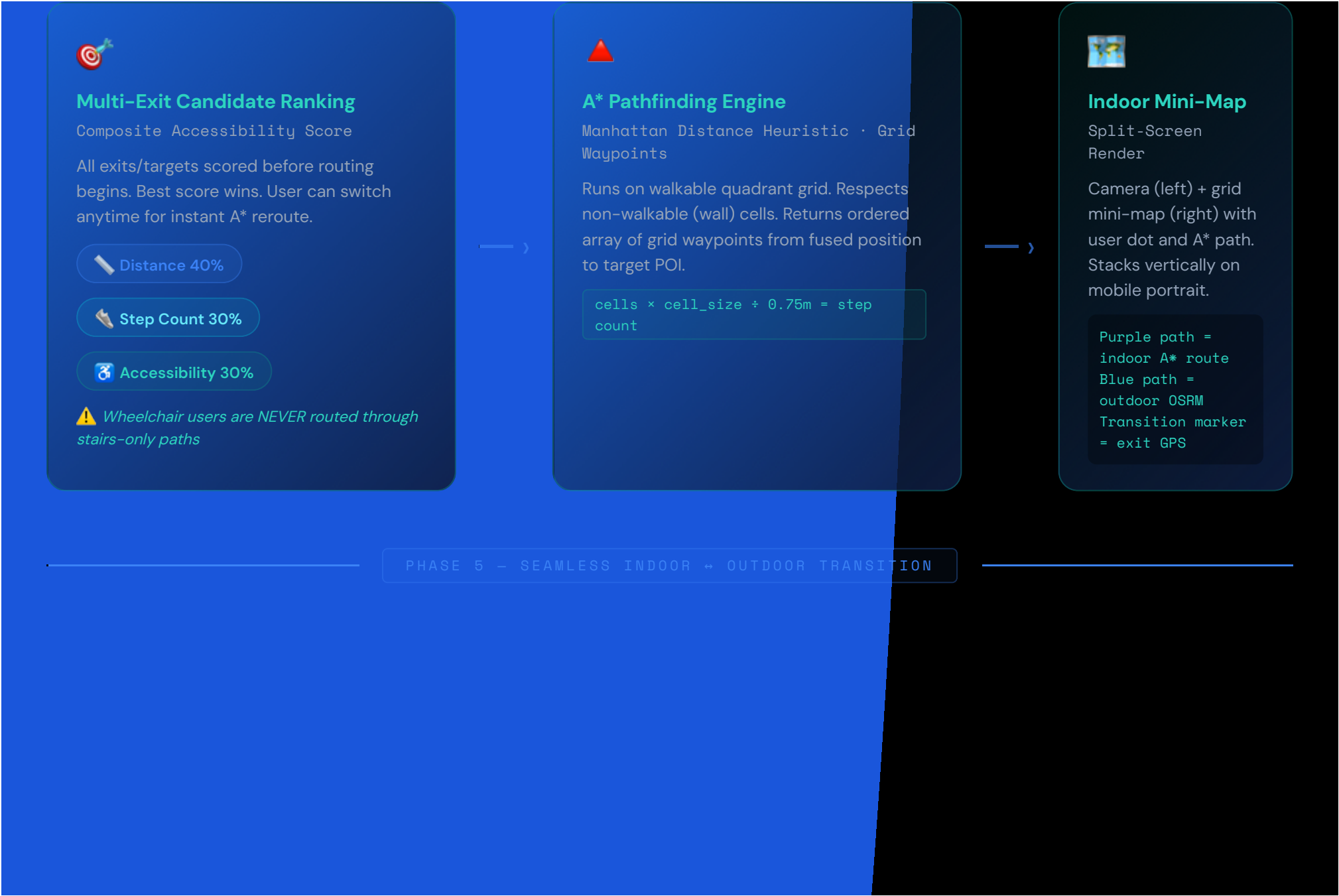
PHASE 1 – USER INPUT

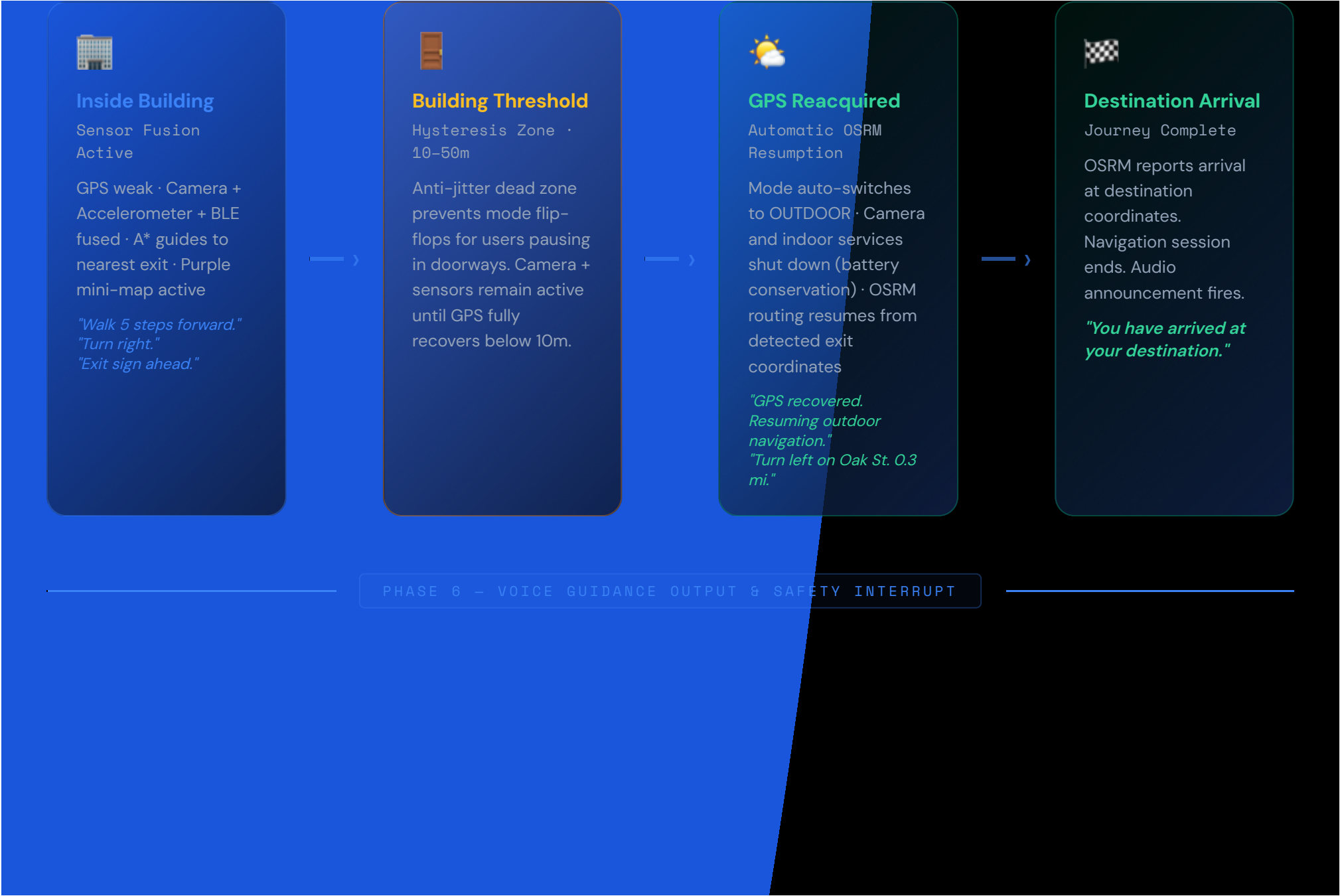




















|                       |                                                       |                                           |
|-----------------------|-------------------------------------------------------|-------------------------------------------|
| DeviceOrientation API | Compass heading for dead reckoning vector             | Direction context for turn instructions   |
| MediaDevices API      | Rear camera at 1280×720 · Client-side only            | Obstacle detection · Fall prevention      |
| OSRM                  | Walking/driving route calculation + turn instructions | Outdoor turn-by-turn voice guidance       |
| Nominatim             | Address → GPS coordinate geocoding                    | Destination resolution from speech        |
| Overpass API          | POI search within 5km radius                          | Find nearby stores, hospitals, pharmacies |
| Web Bluetooth API     | BLE beacon RSSI scanning                              | Indoor trilateration positioning layer    |

285M

Visually Impaired  
Users Worldwide

11

JavaScript Modules

6,800+

Lines of Code

5

Indoor Positioning  
Layers

50+

Voice Synonym  
Normalizations

30%

Accessibility Weight in A\* Scoring

9

External API Integrations

0

Video Frames Sent to Cloud

